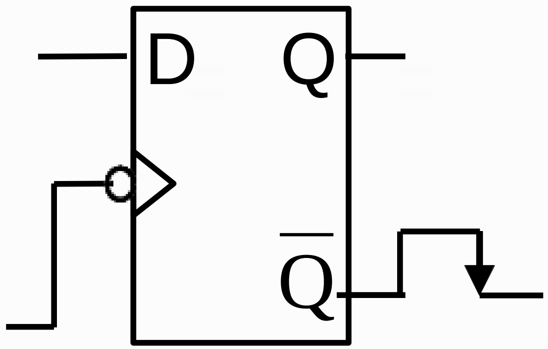
REGISTER

Introduction – Shift Register

- Shift registers are constructed using several flip-flop, connected in such a way to STORE and TRANSFER/ Shift digital data.
- Basically, D flip-flop is used. The input data (either ‘0’ or ‘1’) is applied to the D terminal and the data will be stored at Q during positive/negative-edge transition of the clock pulse.



Shift Register

- One D FF is used to store 1-bit of data. Thus, the number of flip-flops used is the same with the number of bit stored.
- Shift register mean that the data in each FF can be transferred/move to other FF upon edge triggering of the clock signal.
- Four types of data movement in shift register are:-
- Serial in –serial out
- Serial in – parallel out
- Parallel in serial out
- Parallel in – parallel out

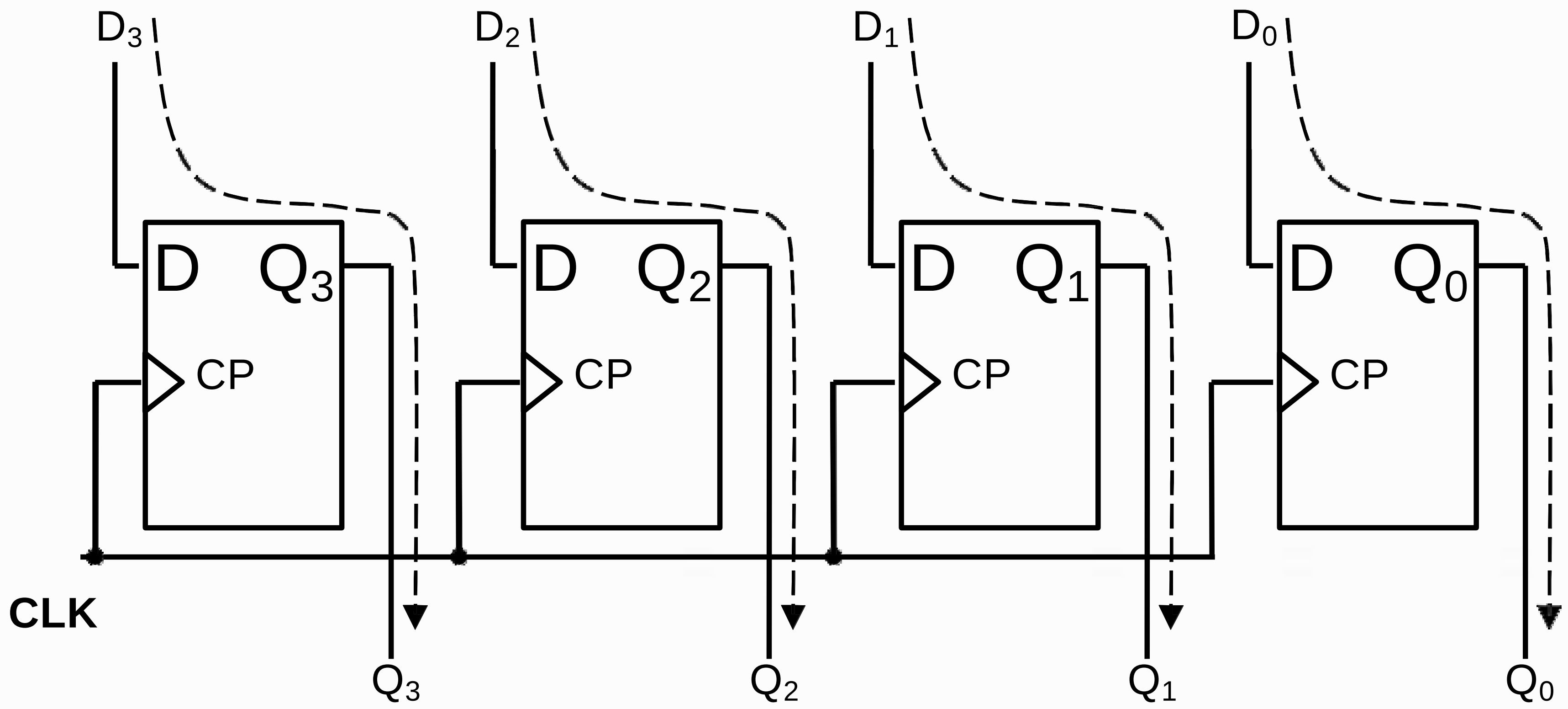
Shift Register

Serial Data Vs Parallel Data movement

Serial	Parallel
•Movement of N-bit data require N number of CLK pulses. Thus, the operation is slow.	•Require only one CLK pulse to transfer all N-bit of data. Thus, operation is faster than serial.
•Only one FF is required to be connected at the output terminal, thus only one wire is required.	•Required N number of connection to the output terminal, which is proportional to the number of bit. Thus, too many connection is required.

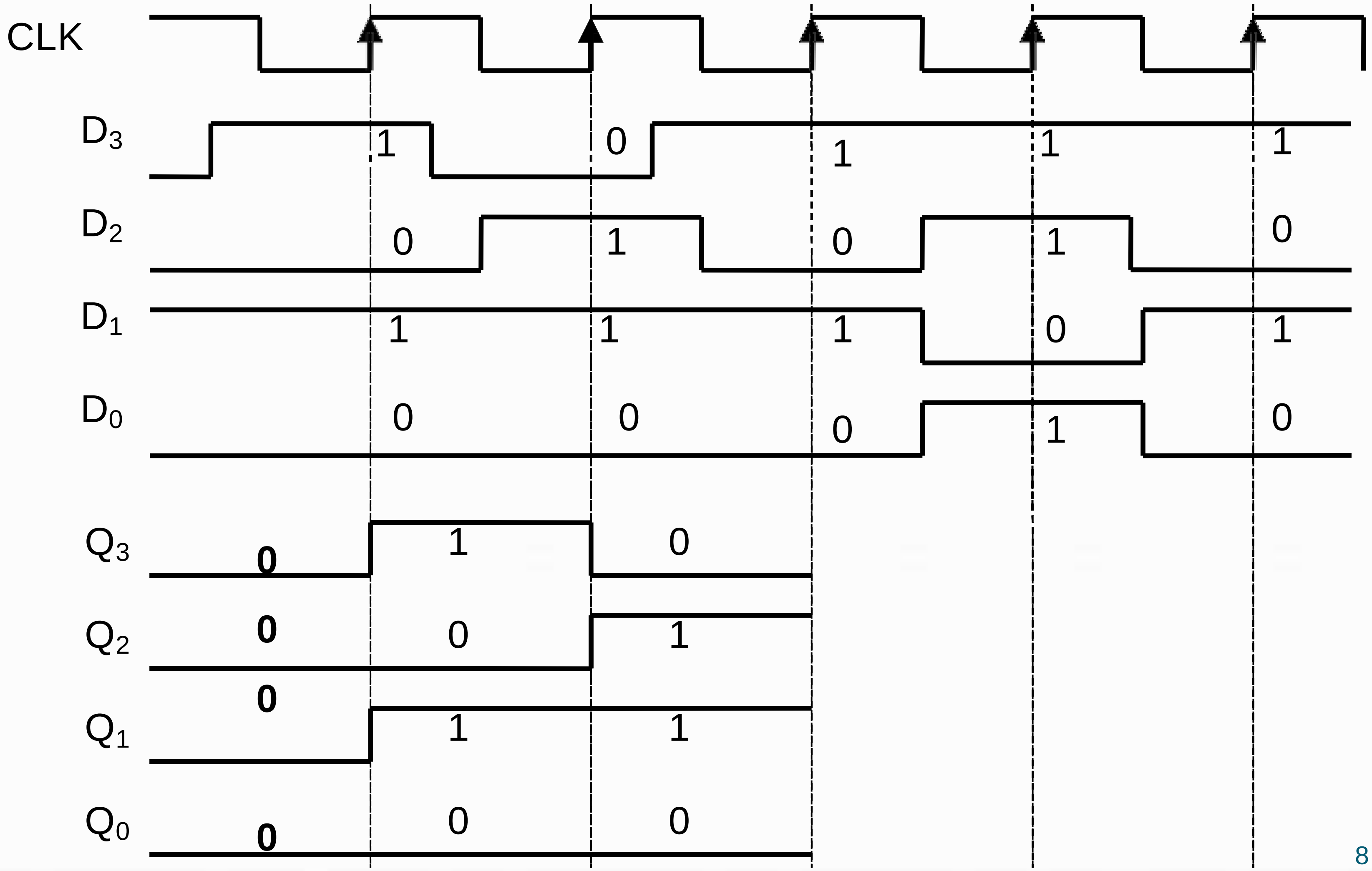
Parallel in / parallel out (PIPO)

- Flip-flop configuration for PIPO register.



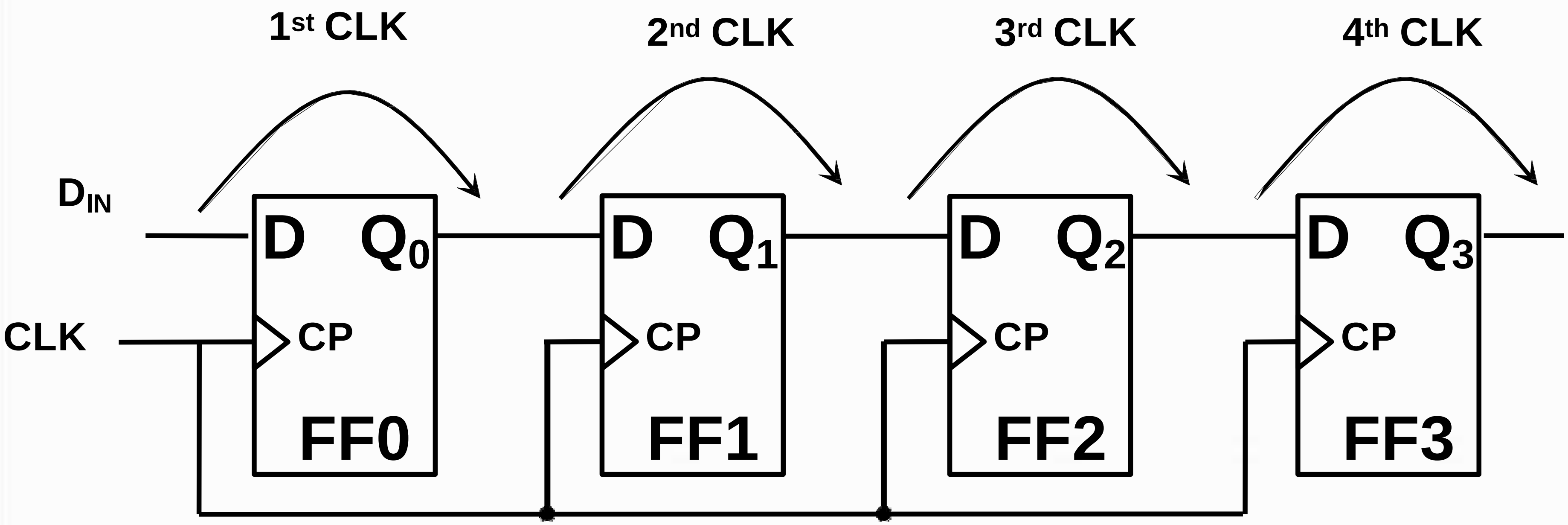
Parallel in –parallel out (PIPO)

PIPO data movement.



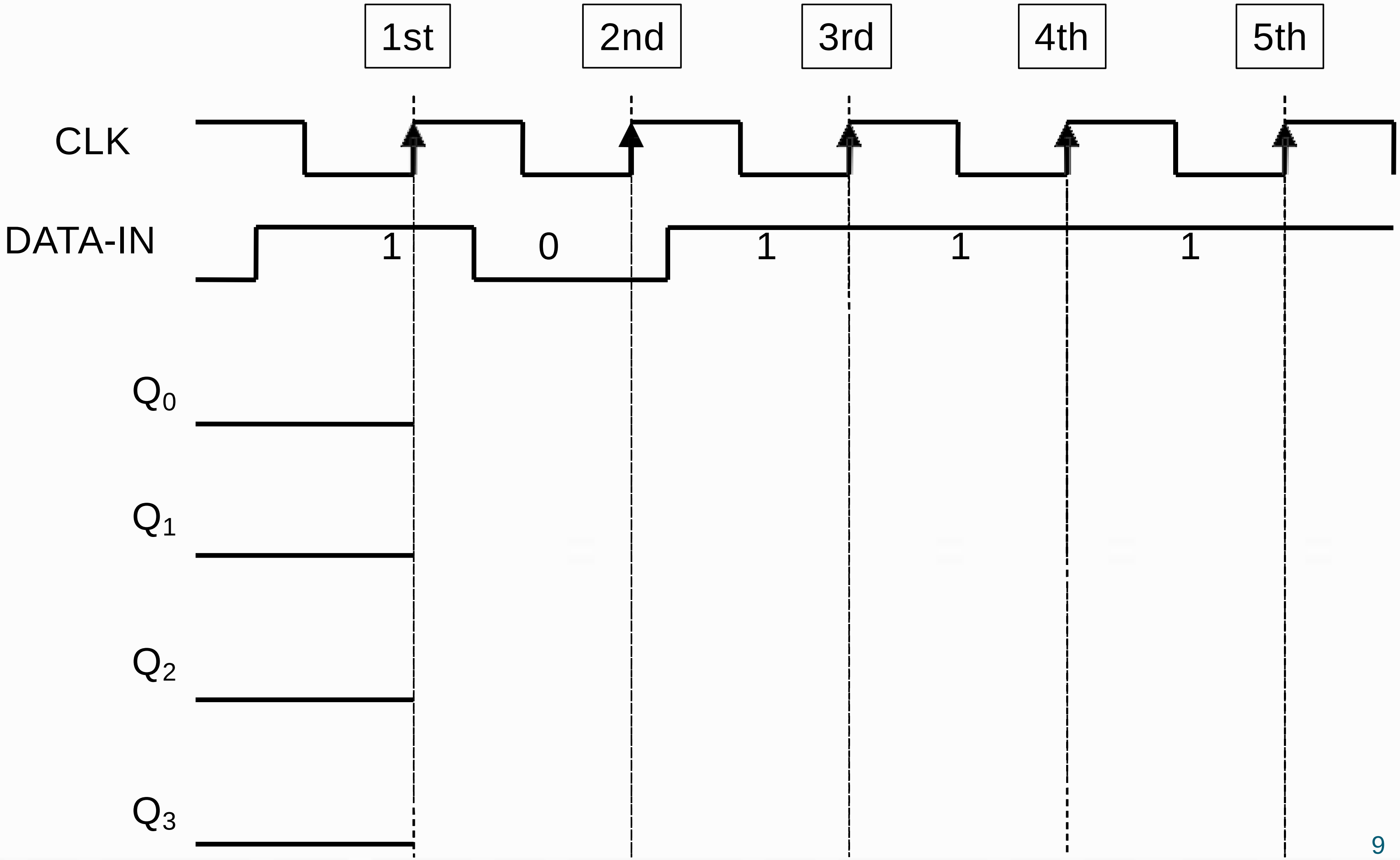
Serial in / serial out (SISO)

- Flip-flop connection for SISO.



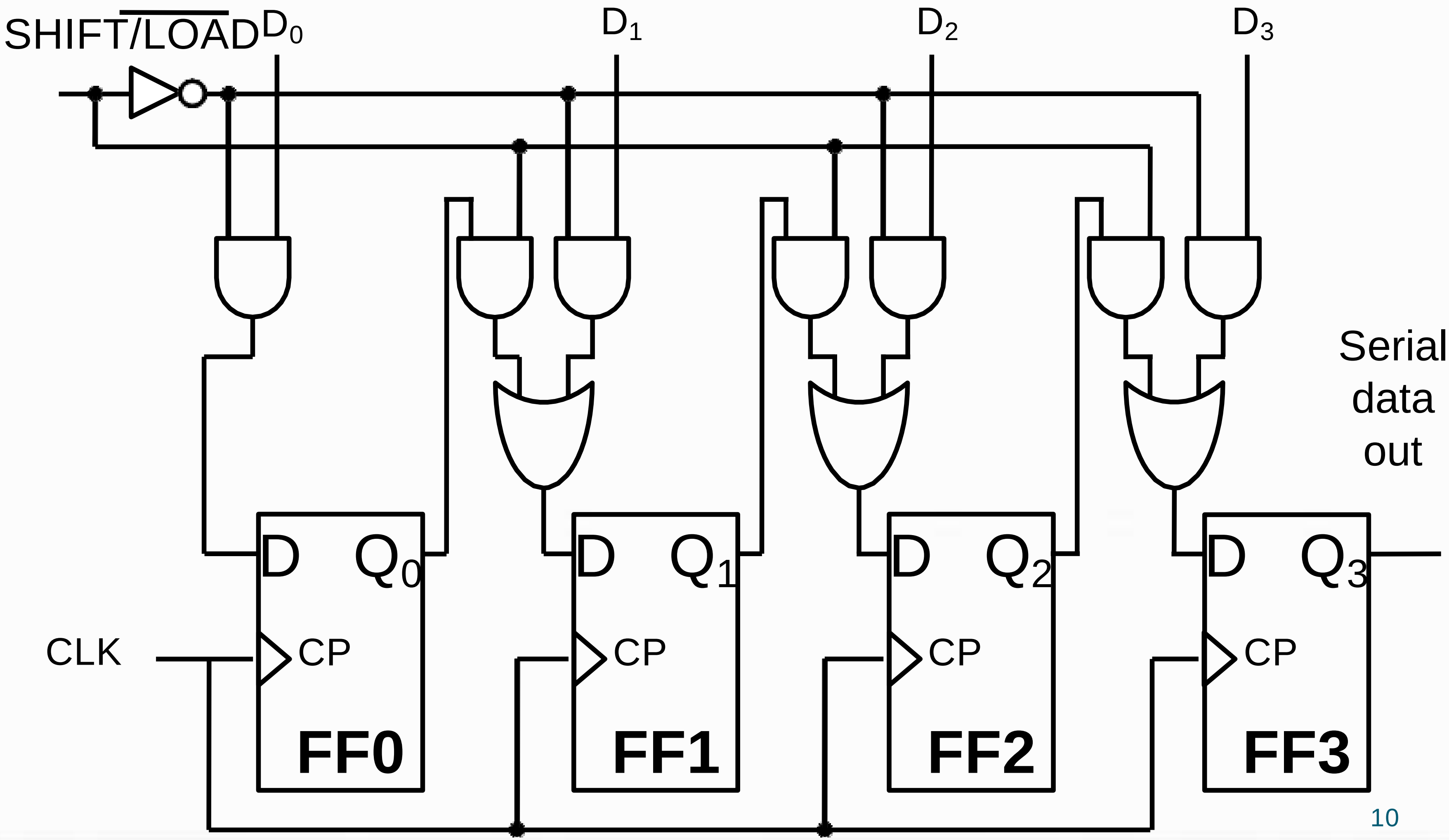
Serial in –serial out (SISO)

SISO data movement. Binary data 10111 is transferred!



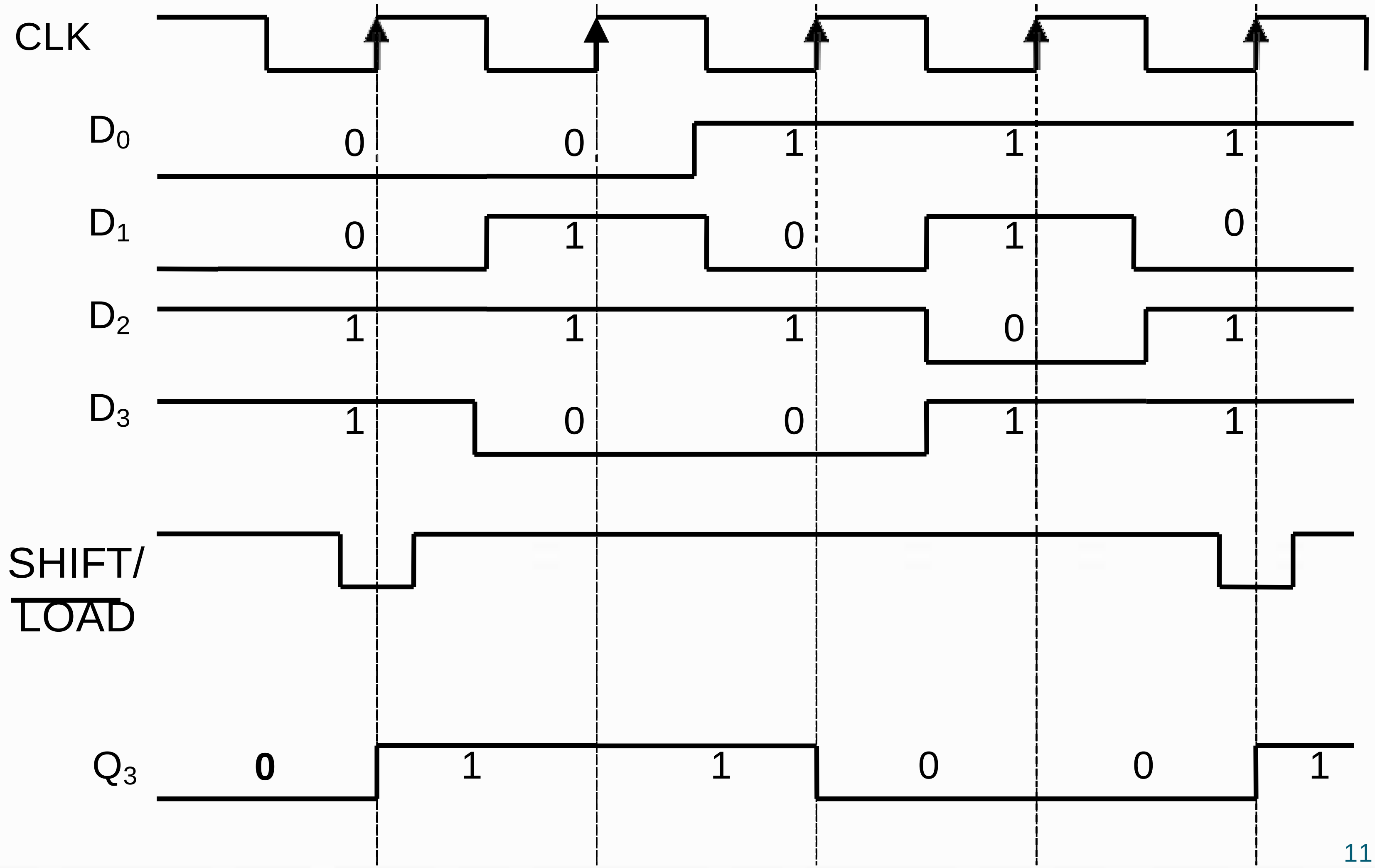
Parallel in / serial out (PISO)

Flip-flop connection for PISO



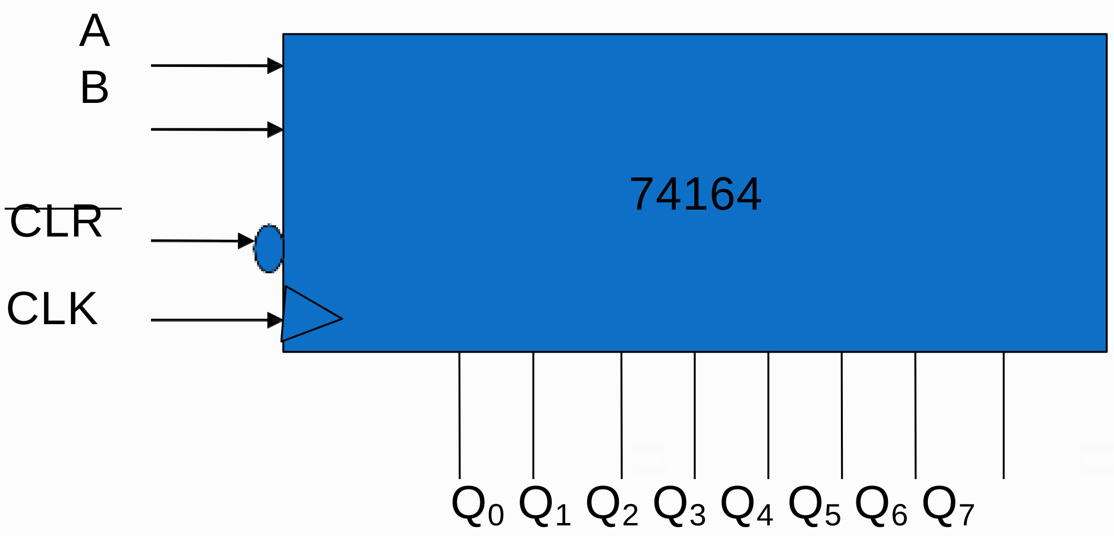
Parallel in/ Serial out (PISO)

- PISO data movement.



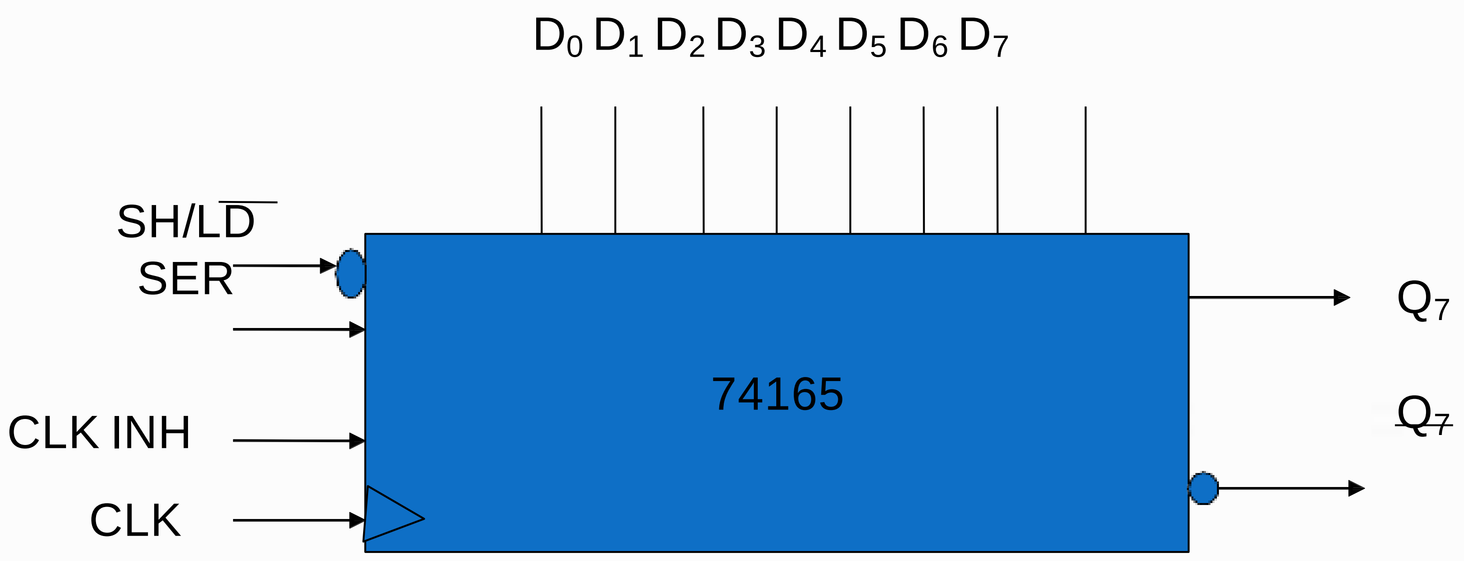
Chips for shift registers

- 74164 is a 8-bit SIPO shift register



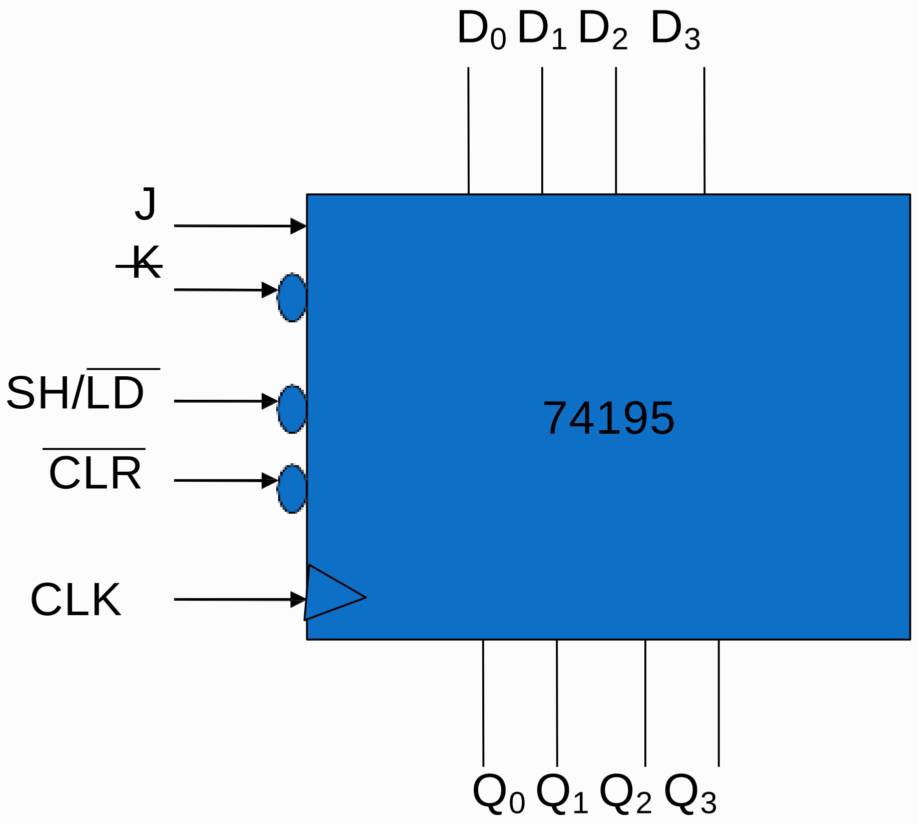
Chips for shift registers

- 74165 is a 8-bit PISO register

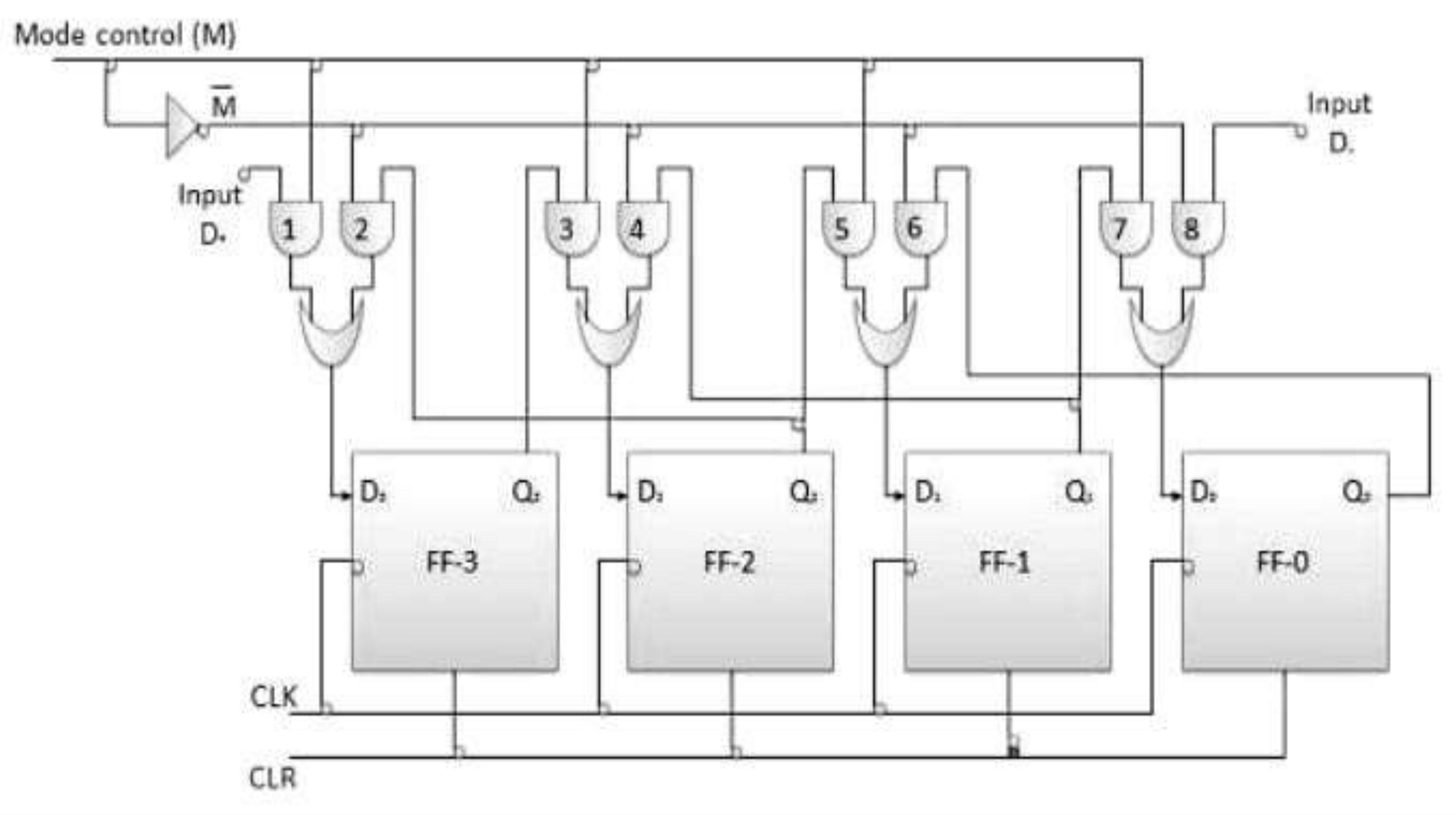


Chips for shift registers

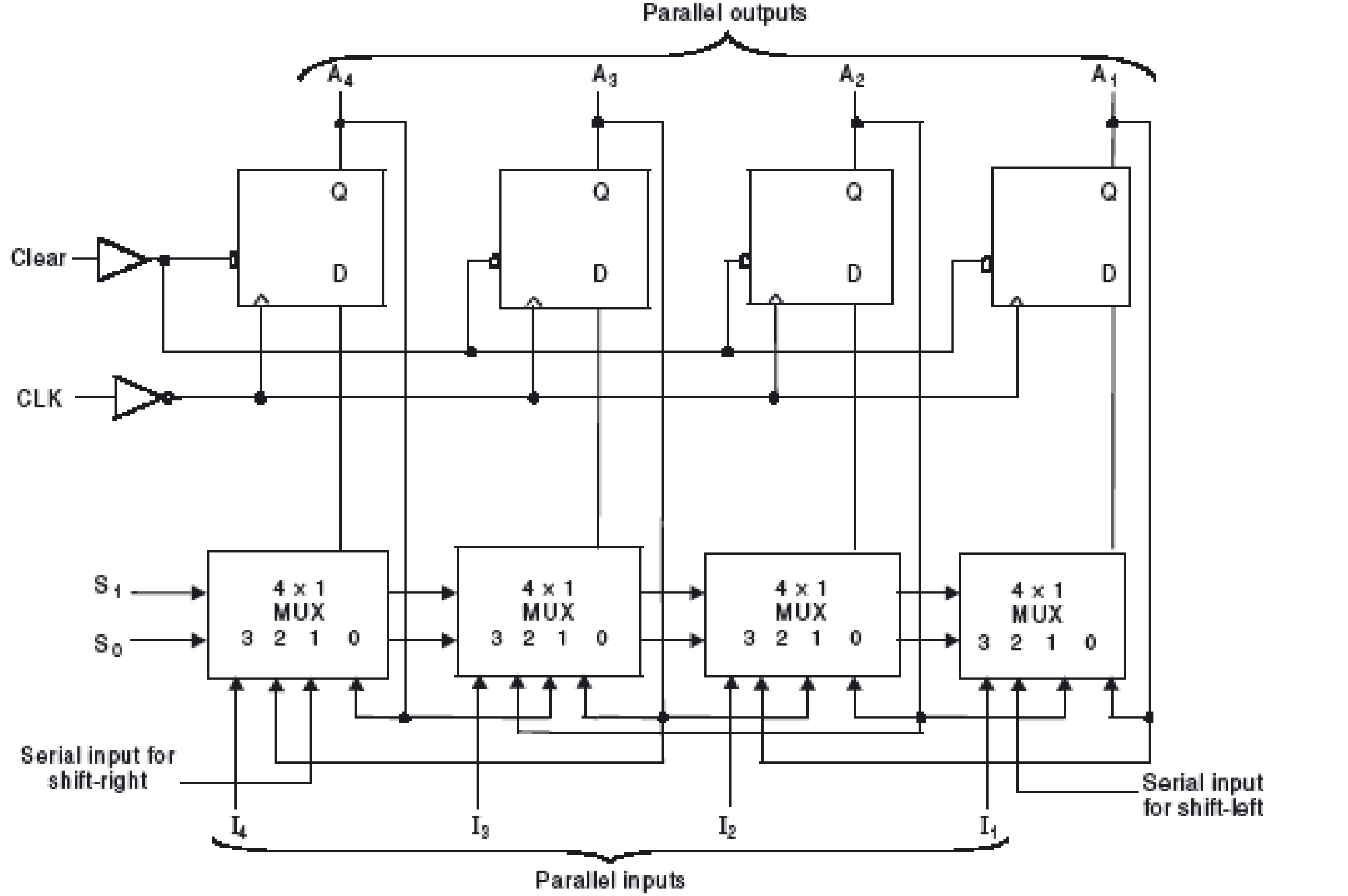
- 74195 can be used as a 4-bit PIPO register



Bidirectional shift register



Universal shift register



4-bit universal shift register.

Shift Register Counters

- A whose output being fed back (connected back) to the serial input. This shift register would count the
  state in a unique sequence!

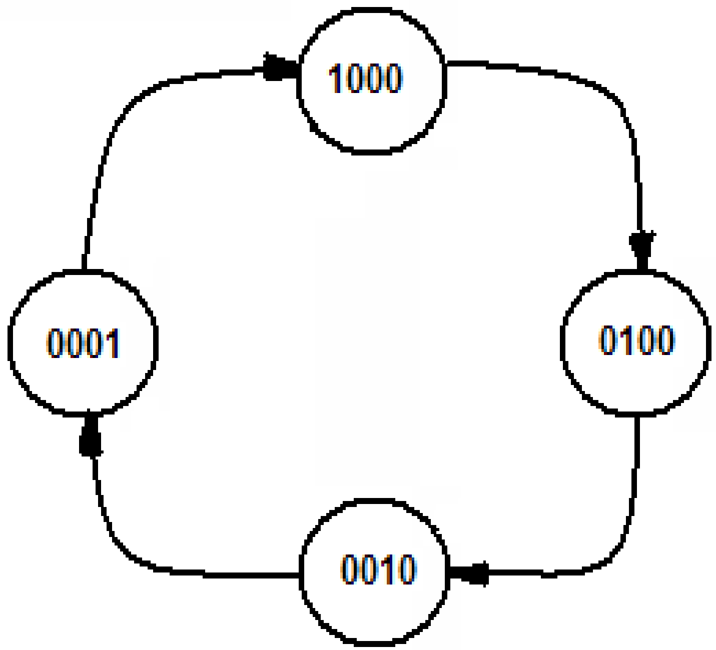
  - Two types of shift register counter:-

    - ***The ring counter***

    - ***The Johnson counter***

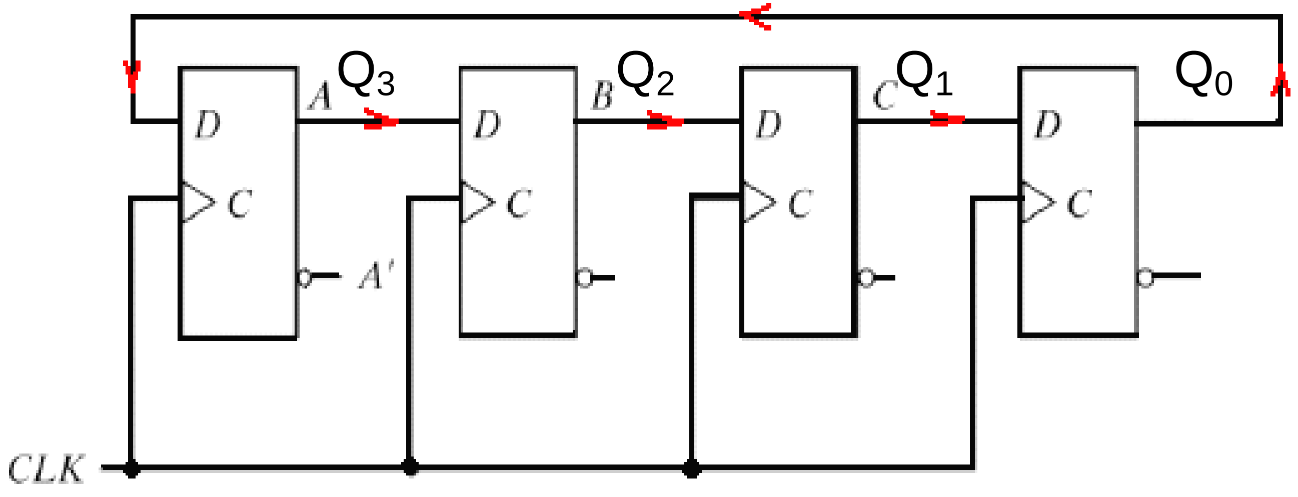
Ring Counter (continue)

- Ring counters are used to construct "One-Hot" counters
- It can be constructed for any desired MOD number
- A MOD-N ring counter uses N flip-flops connected in the arrangement as shown in fig. a)
- In general ring-counter will require more flip-flops than a binary counter for the same MOD number



d) State Diagram

Ring Counter



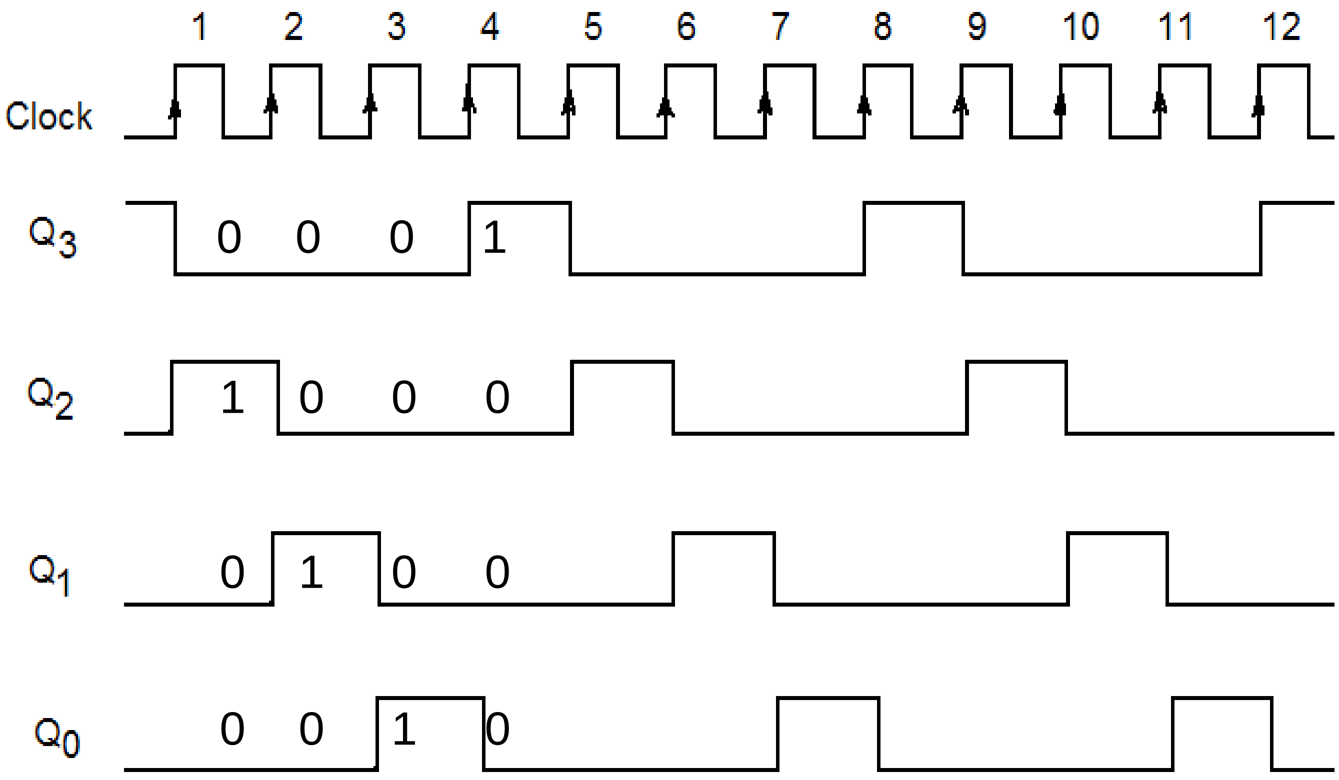
a) Four-bit ring counter

Ring Counter (continue)

Sequence Number	A	B	C	E
	Flip-Flop outputs			
1	1	0	0	0
2	0	1	0	0
3	0	0	1	0
4	0	0	0	1
5	1	0	0	0
6	0	1	0	0
7	0	0	1	0
8	0	0	0	1

c) Sequence table

Ring Counter (continue)



b) Four-bit ring counter waveform

Ring Counter (continue)

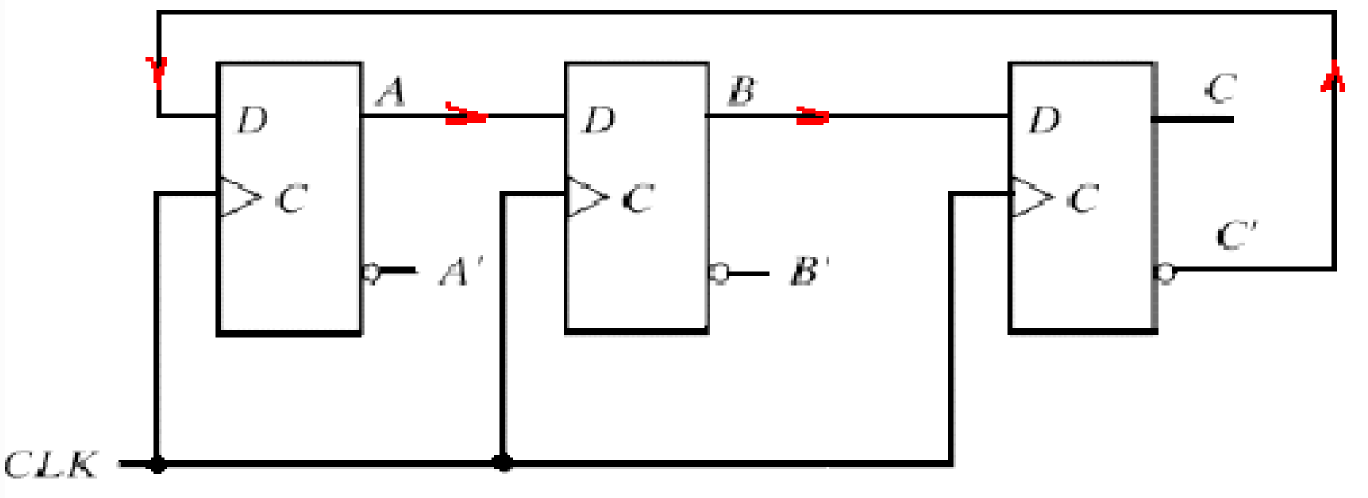
- Exercise: Draw a 3 Bit Ring Counter Circuit with initial input 010. show a True Table until 8 clock pulse/number sequence and draw the output waveform.

- Answer:

  Discuss with Your lecturer

Johnson Counter

Or Twisted-ring counter



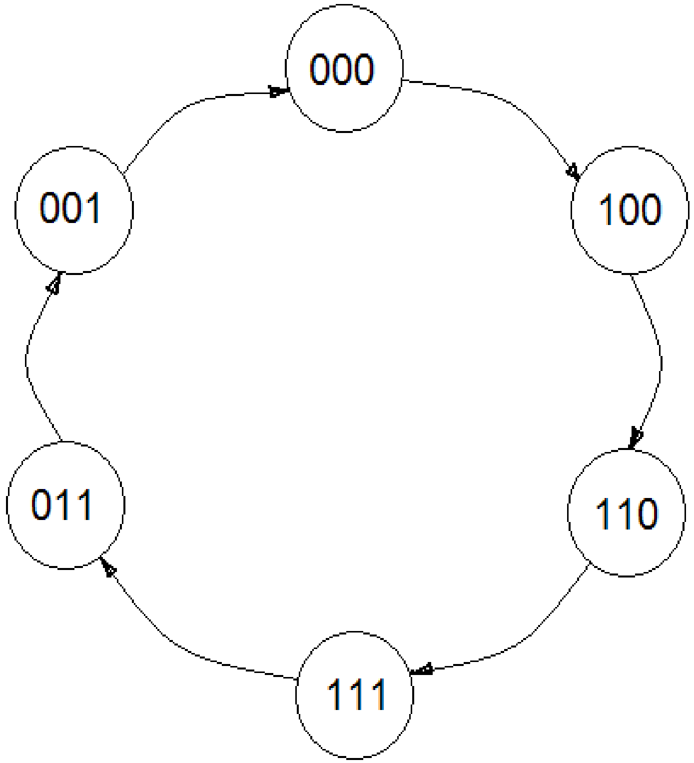
a) MOD-6 Johnson counter

- Johnson counter constructed exactly like a normal ring counter except that the inverted output of the last flip-flop is fed back to first flip-flop

Johnson Counter
(Continue)

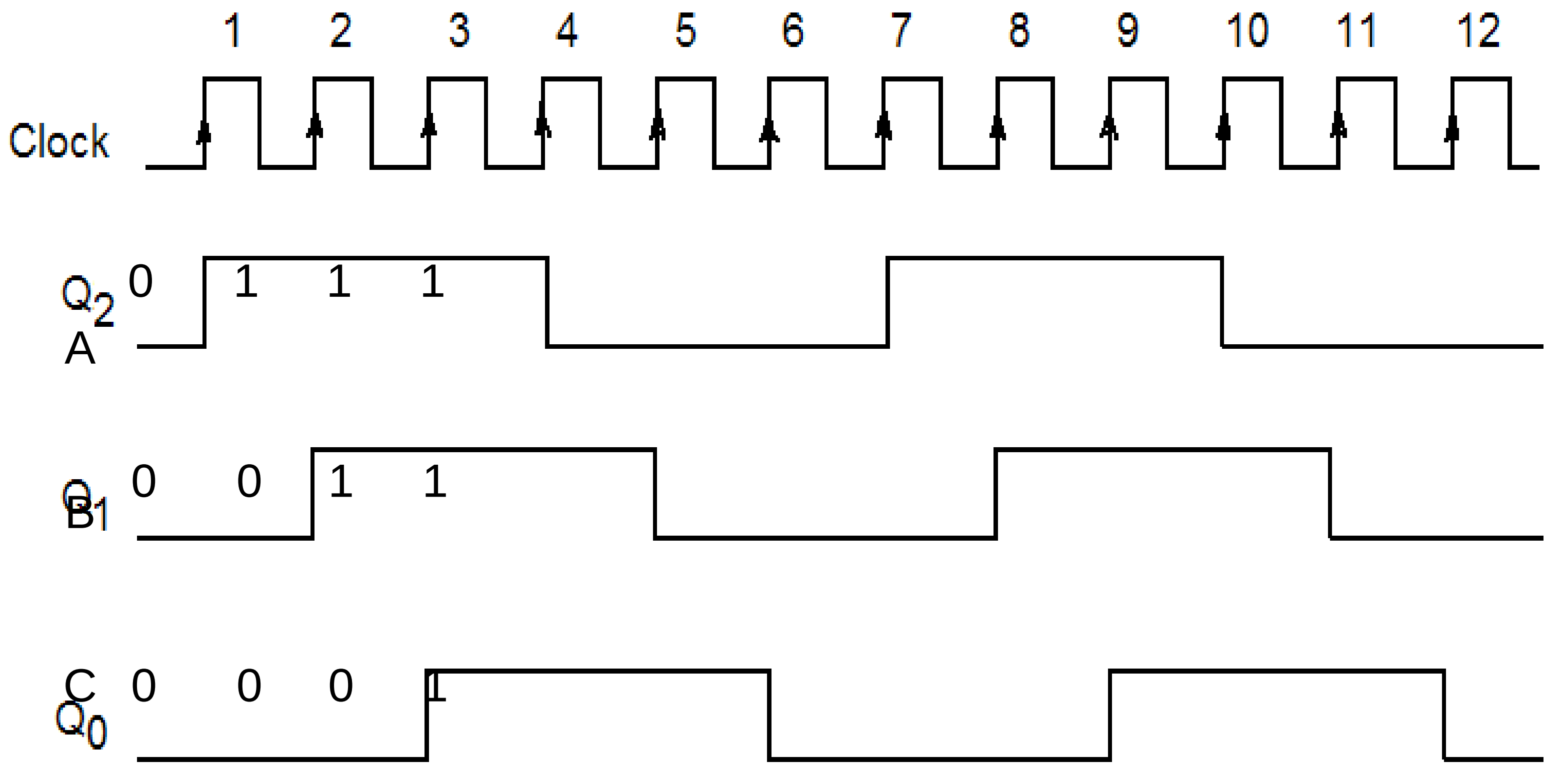
Sequence Number	Flip-Flop outputs A	B	C
0	0	0	0
1	1	0	0
2	1	1	0
3	1	1	1
4	0	1	1
5	0	0	1
6	0	0	0
7	1	0	0
8	1	1	0
.	.	.	.
.	.	.	.
.	.	.	.

c) Johnson Counter sequence table



d) Johnson counter state diagram

Johnson Counter
(Continue)



b) MOD-6 Johnson counter waveform

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1. "Digital Systems Principles And Application"
   Sixth Editon, Ronald J. Tocci.

2. "Digital Systems Fundamentals"
   P.W Chandana Prasad, Lau Siong Hoe,
   Dr. Ashutosh Kumar Singh, Muhammad Suryanata.

End Of This Topic

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